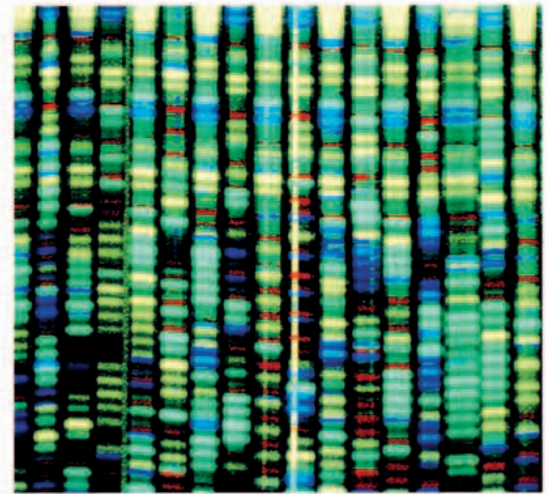


The human genome

You started life as a cluster of cells containing the genetic instructions needed for you to grow into a unique person. Each of the 100 trillion cells in your body contains a copy of your genome—the coded information found in the two sets of 23 chromosomes inherited from your parents. Throughout life, this controls the way your body develops.

THE HUMAN GENOME PROJECT

Each DNA molecule that makes up the genome is like a twisted ladder linked by rungs called bases (see page 164). There are four different types of bases, and the sequence in which they occur in DNA provides the coded instructions—or genes—that make us human. In 1990, scientists began the Human Genome Project to determine the sequence, a goal that they reached in 2003.



SEQUENCING DNA

Decoding the entire human genome means working it out letter by letter—a process known as sequencing. Scientists do this by chopping up the DNA into shorter strands and then separating these out by size to form a series of bands (a banding pattern). The base (A, T, C, or G) at the end of each strand is color-coded by chemicals, and the colored bands are then read by a computer and turned into a sequence of letters.

◀ **GROWING DNA** *This robotic arm is holding a tray containing colonies of bacteria, which are used to grow human DNA. Tiny pieces of DNA are cloned by the bacteria to make amounts large enough for scientists to analyze.*

